Anton Zarubin

Senior Software Engineer

Slot Engines · RTP Modeling · RNG · Simulation · Backend Architecture

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Senior Software Engineer specializing in slot game logic, RTP modeling and deterministic systems. Designed and built production-ready slot engines with full control over probability, payout behavior and simulation-driven balancing. Experienced in scalable backend-oriented architectures, high-performance simulation pipelines and reproducible game logic. Strong expertise in slot mechanics, volatility tuning, RNG systems and real-time multiplayer architectures.

Core Technologies

Slot game development · RNG systems · Probability systems · RTP / volatility modeling · Deterministic systems · Backend architecture · TypeScript · Node.js · Phaser 3 · Game logic architecture · Simulation systems · Config-driven design · CLI tooling · Performance optimization

Gambling Systems Engineering

- Built deterministic slot engines with reproducible outcomes and no hidden state
- Implemented config-driven reels, symbols and payout systems
- Performed RTP and volatility modeling through large-scale simulation
- Designed real-time multi-client architecture (CLI + Web client)
- Separated core game logic from rendering and UI layers
- Implemented full slot lifecycle (bet → spin → outcome → payout)
- Designed probability distribution systems for symbol generation
- Built production-ready architecture focused on scalability and testability

Slot System Features

- Reel-based slot architecture with configurable symbol distribution

- Modular symbol pipeline and payout evaluation system
- Support for multiple RTP/volatility profiles via configuration
- Reproducible simulation runs using deterministic RNG
- Integration-ready core independent from rendering layer
- Scalable simulation execution (10k–100k+ spins per run)
- Full slot lifecycle: bet handling, spin resolution and payout calculation
- Backend-ready core architecture for integration into game services

Impact

- Built deterministic slot engine for reproducible simulation and balancing
- Enabled rapid iteration of reels, payouts and symbol behavior
- Implemented large-scale simulation workflows for RTP validation
- Designed multiplayer architecture supporting multiple clients
- Eliminated hidden state to guarantee reproducibility
- Designed systems with production constraints: scalability and stability
- Built backend-oriented core systems ready for service integration

Projects

- CatSpin Arena (Live Demo) — https://catspin.catemup.com/
- Live playable deterministic slot prototype
- Real-time multiplayer architecture (CLI + Web)
- Full slot lifecycle: bet → spin → resolve → payout
- RTP / volatility simulation system
- Config-driven slot design

Experience

Gambling Systems Engineer

Personal / Showcase Project

Slot Game Logic · RTP Modeling · Simulation · Backend Architecture

TypeScript · Node.js · Phaser 3 · Deterministic core · Simulation CLI · Web client · RNG systems · Probability modeling · Backend-oriented architecture

Built a deterministic slot engine with full control over configuration, probability and payout behavior.

CatSpin Arena — Deterministic Multiplayer Slot Engine

Live playable slot prototype built on deterministic core.

- Implemented deterministic spin logic across all clients
- Designed real-time multi-client architecture (CLI + Web)
- Separated core logic from UI and rendering
- Built config-driven reels and payout pipeline
- Implemented deterministic RNG for reproducible gameplay
- Built and deployed live playable prototype

Slot Implementation (Phaser)

Slot prototypes integrated with deterministic backend.

- Implemented reel generation based on probability distribution
- Built full spin lifecycle (bet → spin → resolve → payout)
- Integrated frontend rendering with deterministic engine
- Designed modular reel system for fast iteration
- Handled animation timing and visual feedback

Simulation & RTP Modeling

Simulation systems for balancing and validation.

- Developed CLI-based simulation system
- Ran large-scale simulations (10k–100k+ spins per run)
- Performed RTP and volatility analysis
- Built strategy testing scenarios
- Validated statistical behavior across configurations

Engineering Approach

Architecture focused on predictability and performance.

- Used config-driven design for rapid balancing
- Eliminated hidden state for reproducibility
- Optimized execution for batch simulation workloads
- Designed reusable core independent from UI
- Built production-ready systems for backend integration